

A Novel Design Methodology for Low-Noise and High-Gain Transimpedance Amplifiers

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Abstract— This paper reports on design and measurement results of a state of the art low-noise and high-gain transimpedance amplifier (TIA) implemented in 0.18 μm TSMC CMOS technology. Thorough design methodology for high gain and low power TIA design for 2.5 Gb/s optical communication circuits family is presented. A noiseless capacitive feedback is proposed and implemented as a noise efficient feedback network for TIA circuits. Besides, analytical noise calculations in this family of TIA circuits are presented and optimum noise criteria are derived. The saturation and instability problem of TIA circuits resulted from DC dark current of the input photodiodes (PDs) is addressed and a circuit level solution is presented. The measurement results of 0.18 μm chip shows bandwidth of 52 kHz to 1.62 GHz, and transimpedance gain of 75.5 dB Ω while dissipating 26.3 mW from a 2.2 V power supply, including the output buffer. Taking advantage of proposed capacitive feedback network and optimum noise criteria, noise measurement results show average input referred current noise of 3.18 pA/ $\sqrt{\text{Hz}}$ for this TIA in the bandwidth of operation.

Index Terms— Current-input circuits, low-noise circuits, optical communications, transimpedance amplifiers

I. INTRODUCTION

RECENTLY, transimpedance amplifiers are being used in a very wide range of applications such as optical communications (families of 2.5, 10, 25,... Gb/s), mechanical sensors, biosensors, and other specific bio-applications like DNA sequencing, and impedance spectroscopy [1-4]. This broad range of applications results in different sets of constraints for a circuit designer when designing a TIA. While in high speed families of optical communication circuits the main goal is pushing the bandwidth of TIA higher in GHz range [4], bio-application TIAs usually stay in much lower frequency ranges but at the same time demand better noise performances and lower power consumption for chronic/long-term implantable versions [1-3]. So, the first question to be addressed in a TIA design is which application

the TIA is being used for and what are the specific design challenges for that specific application.

Generally, as the first building block in a receiving front end, no matter what the application is, TIAs convert the input current to an amplified voltage for use in the next blocks. System performance in terms of bandwidth, sensitivity and signal to noise ratio (SNR) is strongly influenced by its first building block. Hence, having enough bandwidth for the application, high and precise transimpedance gain in the bandwidth of interest, and low input current noise for higher sensitivity while dissipating reasonable power are the fundamental features expected from a well-designed TIA.

In TIA design for 2.5 Gb/s optical communications family, the fact that the bandwidth is still in range of few GHz, leads to higher gain and lower noise, at the same time, to be the most important design challenges. Increasing gain usually increases chance of instability and low noise performance usually translates into higher power consumption. Especially in terms of instability, in addition to typical issues that designers should take into account for a voltage amplifier, such as phase margin and gain margin, saturation and instability resulting from DC component of the input current can add an extra level of difficulty for TIA design [4]. The input current is not ideally a symmetric AC current at the input of TIA, and usually there is a non-ideal DC current component. In optical communication circuits, this DC current comes from mismatch between the two series inverse-biased input photodiodes that are used to sense the optical beams at the end of optical fiber. For each of the mentioned problems, understanding the problem and fundamental limitations is the key issue which can result in finding an optimum solution.

General accepted topology for TIA circuit is a voltage amplifier with shunt-shunt feedback across it, as shown in Fig. 1. This feedback topology is opted because of low input impedance, as desired for a current-input block and low output impedance, as needed for a voltage-output amplifier. This feedback network also assures almost constant transimpedance gain in the bandwidth of interest, while decreasing sensitivity to process and temperature variation.

For the voltage amplifier, designer has plenty of choices. Because the noise performance of a TIA is another important

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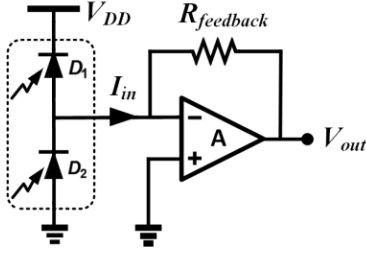


Fig. 1. Typical TIA topology; a voltage amplifier with shunt resistive feedback.

design challenge, the simpler voltage amplifier with less number of active components is opted, the less input current-noise would be resulted. So, a single-stage amplifier can be a viable candidate if enough gain can be achieved. Choice of input topology for voltage amplifier is mostly dictated by noise and bandwidth constraints. Input parasitic capacitances of photodiode, electro-static discharge (ESD) circuit, and input PAD at the input of TIAs primarily constrain their bandwidth and noise performances [6]. As a result, different input topologies have been proposed to relax the effect of input capacitance, such as common gate (CG) [7], regulated cascode (RGC) [6,17], common drain (CD) [8], and common source (CS) [5]. At first glance, CG and RGC topologies seem better candidates in terms of relaxing the effect of input parasitic capacitances, but high current noise contribution of the load and input transistors in these topologies, degrades the noise performance drastically [4]. Isolating property of CD topology can be useful but it cannot be a good choice, first because we still need another stage for amplification, and also high noise contribution of the buffer transistor in addition to amplified (divided by smaller than unity gain of buffer) noise from second stage makes this topology a poor candidate in terms of noise efficiency. So, a voltage amplifier with a CS topology in its input would be the first step in the design procedure.

For choice of feedback network, the most common approach is a resistive feedback network, as shown in Fig. 1. Resistor value doesn't change in the bandwidth of interest and also provides DC current path for the dark current of PDs, although in most cases because of headroom problems another DC path is designed and implemented and this resistor doesn't serve as a DC path for input current [4]. However, the main drawback of resistive feedback is its noise performance. Current noise of this resistor is directly added to the input referred current noise of the TIA which degrades the noise performance drastically [4, 14]. A capacitive, noiseless feedback network can serve as a viable solution for this problem. As shown in Fig. 2, we proposed [5] a TIA with capacitive feedback network for maintaining high gain and low noise at the same time. Also a circuit level solution for the DC dark current in this topology is provided.

The rest of the paper is organized as follows; in section II, the basic TIA topology, analytical calculations of the TIA specifications including gain, band-width, and input referred noise are reviewed. Besides, noise analysis is performed and optimum criteria for noise calculation are derived.

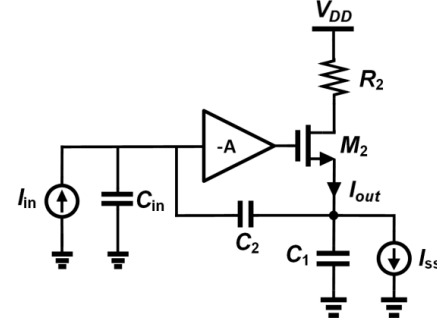


Fig. 2. General Concept of the proposed TIA with capacitive feedback

Also, a novel design methodology is proposed for this TIA family. In section III, circuit structures are shown and the final proposed topology is presented. In section IV, the measurement results of implemented TIA in 0.18 μm CMOS technology is presented and compared with other recent publications. Finally, section V. draws some conclusions from this work.

II. BASIC CONCEPT AND ANALYSIS

A. Basic Concept

In this section, basic principles of the proposed topology are reviewed [5]. For sake of simplicity, detail of the common source input stage is neglected in this section and will be taken into account in next section.

Fig. 2 shows general idea of the capacitive feedback in which $C_{in} = C_{PD} + C_{A,Parasitic}$, and input photodiodes (PDs) are modeled by a current source in parallel with a capacitor.

The transfer function of the proposed topology is calculated to be

$$\frac{I_{out}}{I_{in}} = \frac{\frac{A(C_1 + C_2) + C_2}{(C_{in} + C_2 + AC_2)}}{1 + \frac{s(C_1 C_{in} + C_1 C_2 + C_2 C_{in})}{g_{m2}(C_{in} + C_2 + AC_2)}} \quad (1)$$

Assuming voltage gain of $A \gg 1$, the current gain in low frequencies can be approximated by $(1 + C_1/C_2)$. To have a better insight about the poles and zeroes of the open loop and the feedback coefficient (β), the block diagram representation of the discussed transfer function is shown in Fig. 3.

As discussed in [5], the -3 dB bandwidth of the circuit can be approximated by

$$f_{-3dB} \approx \frac{g_{m2} \cdot (1 + A)}{2\pi \cdot C_1} \quad (2)$$

The capacitive feedback network does not cause any additional noise. Neglecting the effect of the input capacitance, the input referred noise voltage of the feed-forward amplifier (A in Fig. 2), $V_{n,A}$, results in an input noise current of

$$I_{noise,A1} = \frac{V_{n,A} \cdot C_1 \cdot s}{1 + C_1/C_2} ; \text{ (Assuming } C_{in} = 0) \quad (3)$$

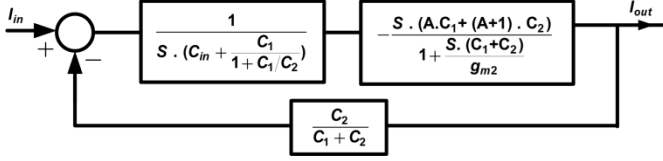


Fig. 3. Block diagram presentation of current transfer function

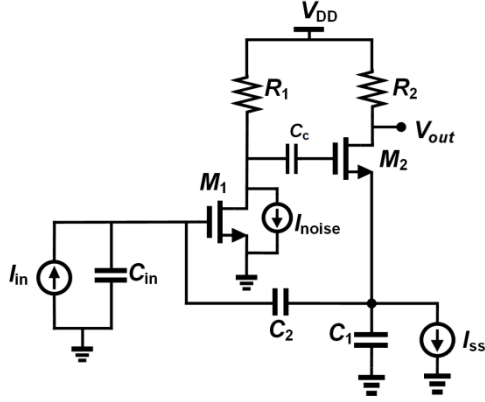


Figure 1. Fig. 4. Basic TIA topology with common source as input stage and capacitive feedback network.

Noise of \$M_2\$ can be represented as a voltage source in series with its gate. This noise would be divided by \$A\$ and added to the noise contribution of the feed-forward amplifier.

Other noise contributors of the circuit are \$I_{ss}\$ and \$R_2\$. As the currents noises of \$I_{ss}\$ and \$R_2\$ are directly added to the output current noise, they would be divided by the current gain \$(1+C_1/C_2)\$ and are referred to the input. As shown in Fig. 4, the common source stage is selected to be used as the feed-forward amplifier due to its superior noise performance.

B. Transimpedance gain

Transimpedance gain of the circuit can be calculated by multiplying the current gain by \$R_2\$

$$\frac{V_{out}}{I_{in}} = \frac{R_2 \cdot \frac{(g_{m1}R_1(C_1 + C_2) + C_2)}{(C_{in} + C_2 + g_{m1}R_1C_2)}}{1 + s \cdot \frac{(C_1C_{in} + C_1C_2 + C_2C_{in})}{g_{m2}(C_{in} + C_2 + g_{m1}R_1C_2)}} \quad (4)$$

where \$g_{m1}R_1\$ is replacing \$A\$, \$C_{in}\$ is sum of the parasitic capacitance of the PDs (\$C_{PD}\$), the gate-source capacitance (\$C_{gs}\$) of \$M_1\$, and the Miller effect of the gate-drain capacitance (\$C_{gd}\$) of \$M_1\$. Eq. 4 can be approximated by \$R_2 \times (1+C_1/C_2)\$ in low frequencies.

C. Input current noise; calculation and optimization

As the current noise of all other noise contributors are suppressed by large gains when referred to the input, the thermal current noise of transistor \$M_1\$ and \$R_1\$ are the dominant noise sources in the TIA. As shown in Fig. 4, \$I_{noise}\$ represents the thermal noise of \$M_1\$ and \$R_1\$. The transfer function from \$I_{noise}\$ to the output node is derived to be:

$$\frac{V_{out}}{I_{noise}} = \frac{\frac{R_1R_2(C_1C_{in} + C_1C_2 + C_2C_{in})}{C_{in} + C_2 + g_{m1}R_1C_2}}{1 + s \cdot \frac{C_1C_{in} + C_1C_2 + C_2C_{in}}{g_{m2}(C_{in} + C_2 + g_{m1}R_1C_2)}} \quad (5)$$

where,

$$C_{in} = C_{PD} + C_{gs} + (1 + g_{m1}R_1)C_{gd} \quad (6)$$

Dividing the noise transfer function by the transimpedance gain, the input referred current noise is

$$\overline{I_{n,in}^2} = \left(\frac{(C_1C_{in} + C_1C_2 + C_2C_{in}) \cdot s}{g_{m1}(C_1 + C_2)} \right)^2 \times \overline{I_{noise}^2} \quad (7)$$

which can be simplified to:

$$I_{n,in} = \left[\left(\frac{C_1}{1 + C_1/C_2} + C_{in} \right) \cdot s \right] \times \frac{I_{noise}}{g_{m1}} \quad (8)$$

As we know:

$$V_{n,A} = \left(\frac{I_{noise}}{g_{m1}} \right) \quad (9)$$

So finally the following expression is obtained for \$I_{n,in}\$:

$$I_{n,in} = \left[\left(\frac{C_1}{1 + C_1/C_2} + C_{in} \right) \cdot s \right] \times V_{n,A} \quad (10)$$

The above equation shows that the previously calculated input current noise in Eq. 3 is only correct when \$C_{in}=0\$. In most practical cases, \$C_{in}\$ in Eq. 6 cannot be neglected as it presents the effect of the capacitances of the large input transistor and photodiode parasitic capacitance. The noise voltage in Eq. 10 is:

$$\overline{V_{n,A}^2} = \frac{4KT\gamma}{g_{m1}} \cdot \left(1 + \frac{1}{g_{m1}R_1} \right) \quad (11)$$

where \$K\$ is the Boltzmann constant, \$T\$ is the absolute temperature, and \$\gamma\$ is the noise excess factor. By substituting voltage noise in Eq. 11 in the current input referred noise in Eq. 10 we would have:

$$\overline{I_{n,in}^2} = \frac{4KT\gamma}{g_{m1}} \cdot \left(1 + \frac{1}{g_{m1}R_1} \right) \cdot \left[\left(\frac{C_1}{1 + C_1/C_2} + C_{in} \right) \cdot s \right]^2 \quad (12)$$

To improve noise performance, the magnitude of the resistor is increased considering the headroom limitation. Thus, the term \$\frac{1}{g_{m1}R_1}\$ can be neglected as it would be much smaller than unity and therefore:

$$\overline{I_{n,in}^2} = \frac{4KT\gamma}{g_{m1}} \cdot \left[\left(\frac{C_1}{1 + C_1/C_2} + C_{in} \right) \cdot s \right]^2 \quad (13)$$

As we know the relation between transconductance and other parameters of the transistor is:

$$g_m = 2\pi \cdot (f_T \cdot C_{gs}) \quad (14)$$

where \$f_T\$ is the cut-off frequency. Assuming that \$M_1\$ is biased to achieve the maximum available \$f_T\$ of the technology, \$g_m\$ of

M_1 is proportional to its C_{gs} according to Eq. 14. Thus (by writing $C_{in} = C_{gs} + C_{PD} + C_{ESD}$ and ignoring C_{gd})

$$\overline{I_{n,in}^2} \propto \frac{\left(C_{gs} + C_{PD} + C_{ESD} + \frac{C_1}{1 + C_1/C_2}\right)^2}{g_m} \quad (15)$$

$$\overline{I_{n,in}^2} \propto \frac{\left(C_{gs} + C_{PD} + C_{ESD} + \frac{C_1}{1 + C_1/C_2}\right)^2}{C_{gs}} \quad (16)$$

which can be minimized if

$$C_{gs} = C_{PD} + C_{ESD} + \frac{C_1}{1 + C_1/C_2} \quad (16)$$

The procedure below can be used to obtain the component values of the circuit considering the power constraint for each stage. C_1 can be calculated from Eq. 2 as g_{m2} is determined by the power budget of the second stage. C_2 and R_2 can be chosen from the required transimpedance gain in Eq. 4. Finally, the aspect ratio (W/L) of M_1 can be calculated considering the optimization condition of Eq. 16. In this design $R_2=570 \Omega$ and $C_1=0.5$ pF were chosen.

III. FINAL TOPOLOGY

In previous section, overall topology was described and design guidelines were reviewed. In this section, based on the reviewed knowledge, the topology is optimized in the circuit level [5]. To maximize R_1 , a PMOS current source (M_P in Fig. 5(a)) is applied to the drain of M_1 . M_P is biased at the maximum possible gate-source voltage to minimize its thermal current noise. As explained earlier, the Miller effect of C_{gd} is a part of the parasitic input capacitance, and due to the high loop-gain required, its effect is significant. Thus, to further improve the noise performance by reducing the miller effect of C_{gd} , using a cascode input stage would be advantageous.

To address the output matching of the TIA, a source follower is added to the second stage which results in a S_{22} of better than -18 dB in the frequency range of interest. The final circuit topology is presented in Fig. 5(a). For low-end cut-off frequency of 50 kHz required in most optical applications, very large resistors are needed in the gate of M_1 and M_2 . To implement such large resistors the bias network suggested in [9] is used as shown in Fig. 5(b).

Regarding the DC dark current, which in this case was due to the mismatches between the two input photodiodes (D_1 and D_2), different approaches has been made by designers. In some cases off-chip methods were proposed to reduce the effect of DC input current [9]. Using the resistor in the feedback loop has also been a solution in cases of resistive feedback [8]. But, here, we propose an on chip, circuit level solution. To alleviate the DC dark current problem, a set of NMOS (as a current source) and PMOS (as a current sink) transistors have been added to the input of the proposed circuit, as shown in Fig. 5(c). In the absence of DC component in the input current, both added transistors are off.

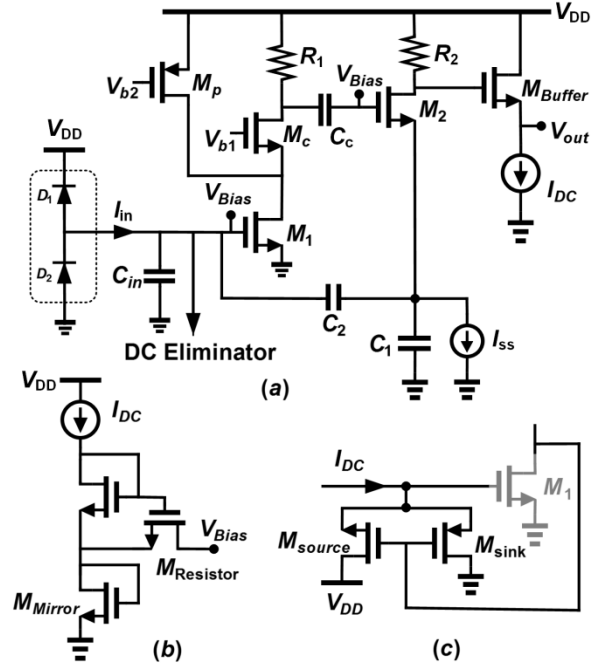


Fig. 5. (a) Main Circuit (b) Bias Network of M_1 and M_2 (c) Proposed DC Eliminator Circuit.

In the presence of a positive DC input current, the gate voltage of M_1 increases and its drain voltage decreases simultaneously, which turns the PMOS transistor on to sink the extra current to the ground.

Similarly, a negative DC input current can be sourced through the NMOS transistor. With accurate design and optimization, the DC bias of the input stage has changed less than 6% in presence of 200 μA of DC current in post-layout simulation, which shows the effectiveness of the proposed method in eliminating the DC component of the input signal.

In the case of an extra DC current, only one of the added transistors is on for a short period of time. Thus, these two transistors add negligible average current noise to the input of the circuit, in case of their operation. Besides, the added transistors are designed with minimum possible widths to minimize their thermal current noise.

IV. MEASUREMENT PROCEDURE AND RESULTS

Based on the design methodology described, a TIA is fabricated in 0.18 μm TSMC CMOS technology. Fig. 6 shows the chip micrograph of the implemented TIA which occupies $400 \times 300 \mu m^2$ of silicon area.

In order to measure specifications of a TIA, different approaches can be taken. A common approach is measuring the S-Parameters of the circuit, just like a LNA, and then calculating the TIA's parameters by modified formulas. This approach is widely used because almost all the measurement set-ups are made for S-Parameters measurement and new instruments would not be needed. The measured S-Parameters of the circuit are shown in Fig. 7.

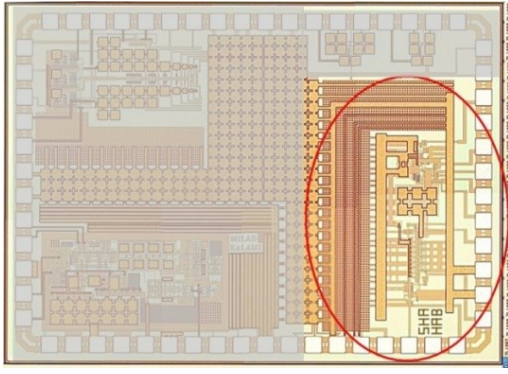


Fig. 6. The chip micrograph of implemented TIA in 0.18 μm technology

In order to calculate the transimpedance gain of a TIA from the shown S-parameters, the formula in Eq. 17 can be used [10].

$$Z_t = \frac{2 \cdot S_{21}}{(1 - S_{11}) \cdot (1 - S_{22}) - S_{21} S_{12}} \cdot Z_0 \quad (17)$$

This formula can be used only when the input of the TIA is directly connected to the network analyzer, and there is no input matching. By using the above formula and measured S-parameters, the transimpedance gain of the circuit is depicted in Fig. 8 and is compared with the simulation results.

The post-layout simulated transimpedance gain was a flat gain of 78 dB Ω with -3dB bandwidth between 52 kHz to 1.81 GHz, while as it can be seen in Fig. 8, the transimpedance gain in the measurement has been decreased less than 3 dB and was measured to be 75.5 dB Ω . Also, bandwidth of the circuit has decreased to 1.62 GHz. The peaking in the frequency response is less than 1.3 dB.

For noise measurement, further considerations must be taken into account. As the noise performance of the circuit can be degraded drastically by the ambient noise, a shielding Faraday cage must be used and the circuit must be installed properly in it. As the input must be opened while measuring the output voltage noise a “not connected” connection must be used at the input to avoid the input acting like an antenna, and prevent instability problems.

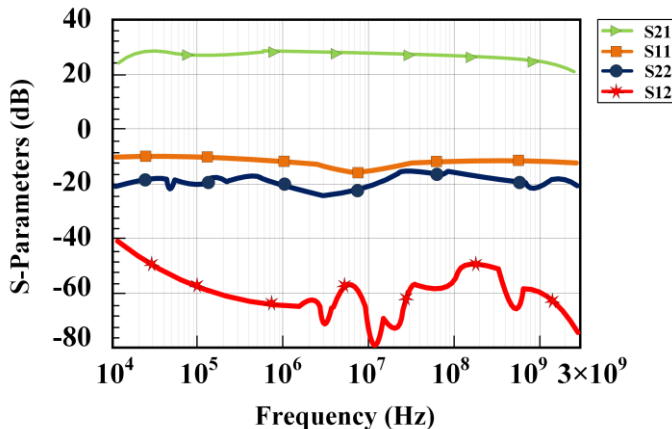


Fig. 7. Measured S-parameters of 0.18 μm TIA chip vs. frequency

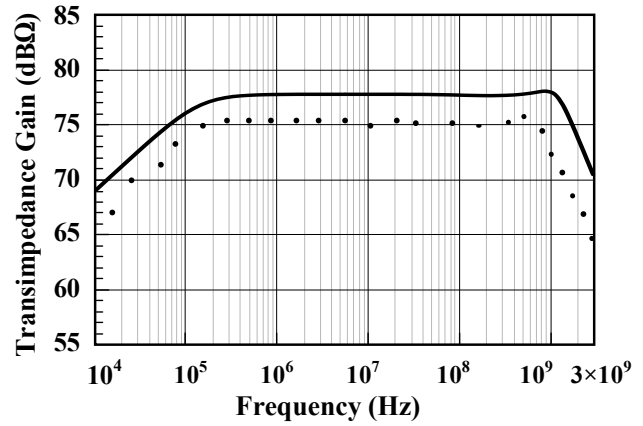


Fig. 8. Measured and post-layout simulated results of transimpedance gain for 0.18 μm TIA chip vs. frequency

Besides, the output noise voltage of the TIA might be lower than noise floor of the common spectrum analyzers, so extra amplification might be needed. In that case, after amplifying the output noise, the noise figure of the added stages must be taken into account.

Considering the above explanation, the final noise measurement result is shown in the Fig. 9 and is compared with the post layout simulation results.

The post-layout simulation results predicted input referred noise current of 1.64 pA/ $\sqrt{\text{Hz}}$ at 1 GHz and an average of 2.1 pA/ $\sqrt{\text{Hz}}$ within the bandwidth of the circuit. Measurement results shows an average input referred noise current of 3.18 pA/ $\sqrt{\text{Hz}}$ within the frequency bandwidth of the transimpedance amplifier.

Regarding the proposed DC eliminating circuit, in test bench characterization, applying DC currents with the amplitudes of up to 10 μA , did not change the output voltage of the TIA. In addition to that, the DC dark current mismatch of photodiodes in real experiment, which was close to 120 nA, did not cause saturation of TIA.

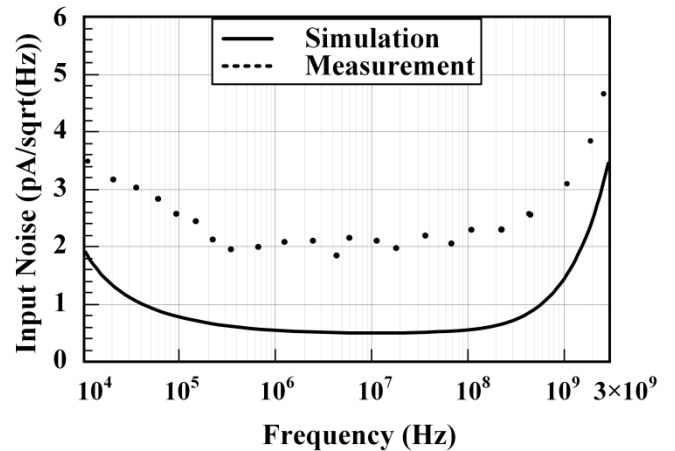


Fig. 9. Measured input referred current noise of the TIA and post-layout simulation results of 0.18 μm Technology vs. frequency.

TABLE I. Measurement and post-layout simulation results comparison with similar recent works.

	This Work Measured / Simulated	[9]	[7]	[11]	[12]	[13]	[15]	[16]	[17]
Technology	0.18 μm CMOS	0.6 μm CMOS	0.5 μm CMOS	0.25 μm CMOS	0.35 μm CMOS	0.18 μm CMOS	0.25 μm CMOS	40 nm CMOS	0.18 μm CMOS
TI Gain (dB Ω)	75.5 / 78	78	64	48	63.5	68	60.1	79.5	71.5
-3dB Bandwidth (GHz)	1.62 / 1.8	0.55	1.2	1.5	1.25	1.73	2.5	1.5	2.21
Power (mW)	26.3 / 26	30	20	72	40	NA	16	4.5	10
Avg. Input current Noise (pA/ $\sqrt{\text{Hz}}$)	3.18 / 2.1	4.5	15	9	16	3.3	*18.9	**8.85	*8.3

*Simulation results **Calculated by authors

In table I, the measurement and post-layout simulation results are compared with results of other recent state of the art publications.

V. CONCLUSION

In this paper, a state of the art low-noise and high-gain transimpedance amplifier for 2.5 Gb/s optical communication circuits family was presented. A thorough design methodology for low noise and high gain TIA design was proposed, analytical noise optimization was performed, and optimum noise criteria were derived. A novel circuit level solution for dark current of input photodiodes were proposed and implemented. Measurement results of the chip, implemented in 0.18 μm TSMC CMOS technology, were shown and their conformity with analytical calculations and analysis were presented.

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REFERENCES

- [1] D. Branton et al., "The potential and challenges of nanopore sequencing," *Nature Biotechnol.*, vol. 26, pp. 1146–1153, 2008.
- [2] Crescentini, M.; Bennati, M.; Carminati, M.; Tartagni, M., "Noise Limits of CMOS Current Interfaces for Biosensors: A Review," *IEEE Transactions on Biomedical Circuits and Systems*, vol.8, no.2, pp.278-292, April, 2014.
- [3] D. Kim; B. Goldstein.; Wei Tang; F.J. Sigworth; E. Culurciello, "Noise Analysis and Performance Comparison of Low Current Measurement Systems for Biomedical Applications" *IEEE Transactions on Biomedical Circuits and Systems*, vol.7, no.1, pp.52-62, Feb. 2013.
- [4] Razavi, B; *Design of Integrated Circuits for Optical Communications*, 2nd edition, New Jersey: Wiley 2012.
- [5] Shahdoost, S.; Medi, A.; Saniei, N., "A 1.93 pA/ $\sqrt{\text{Hz}}$ transimpedance amplifier for 2.5 Gb/s optical communications," *In Proc. Of IEEE International Symp. on Circuits and Systems (ISCAS)*, pp.2889-2892, 15-18 May 2011.
- [6] S. M. Rezaul Hasan "Design of a Low-Power 3.5–GHz Broad-Band CMOS Transimpedance Amplifier for Optical Transceivers", *IEEE Trans. on Circuits and Systems: Part I. Regular papers*, vol. 52, No. 6, pp. 1061-1072, June 2005.
- [7] C. Toumazou and S. M. Park, "Wideband low noise CMOS transimpedance amplifier for gigahertz operation," *Electron. Lett.*, vol. 32, no. 13, pp. 1194–1196, Jun. 1996.
- [8] T. Yoon and B. Jalali, "1 Gbit/s fiber channel CMOS transimpedance amplifier," *Electron. Lett.*, vol. 33, no. 7, pp. 588–589, Mar. 1997.
- [9] B. Razavi, "A 622 Mb/s 4.5 pA/ $\sqrt{\text{Hz}}$ CMOS transimpedance amplifier," *In Dig. Tech. Papers of IEEE Int. Solid State Circuit Conf. (ISSCC)*, pp. 162–163, Feb. 2000.
- [10] Jun-De Jin; Hsu, S.S.H., "40-Gb/s Transimpedance Amplifier in 0.18- μm CMOS Technology," *Proceedings of the 32nd European Solid-State Circuits Conference, ESSCIRC 2006*, vol. 8, no. 3, pp.520-523, 19-21 Sept. 2006.
- [11] Mitran, P., Beadoin, F., and El-Gamal, M.N.: "A 2.5 Gbit/s CMOS optical receiver frontend". *Proc. of IEEE Int. Symp. on Circuits and Systems, ISCAS*, Vol. 5, pp. 441-444, 2002.
- [12] K. Schrödinger, J. Stimma, and M. Mauthe, "A fully integrated CMOS receiver front-end for optic gigabit ethernet," *IEEE, Journal of Solid State Circuits*, vol. 37, pp. 874–880, July 2002.
- [13] Yong-Hun Oh, Sang Gug Lee, "Inductance Enhancement technique and its application to a Shunt-peaked 2.5 Gb/s transimpedance amplifier design", *IEEE Trans. on Circuits and Systems: Part II: express briefs*, pp. 624-628, 2004.
- [14] Atef, M.; Zimmermann, H., "Optical Receiver Using Noise Cancelling With an Integrated Photodiode in 40 nm CMOS Technology," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol.60, no.7, pp.1929-1936, July 2013.
- [15] Seung-Hoon Kim; Chi-Uk Kam; Jong-Hwa Lee, "Design of 2.5Gb/s Transimpedance Amplifier using CMOS Technologies," *In Proc. Of 9th Int. Conf. on Advanced Communication Technology*, vol.3, no.2, pp.1825-1828, 12-14 Feb. 2007.
- [16] Atef, Mohamed; Zimmermann, H., "2.5Gbit/s transimpedance amplifier using noise cancelling for optical receivers," *In Proc. Of IEEE International Symposium on Circuits and Systems (ISCAS)*, vol.3, no.2, pp.1740-1743, 20-23 May 2012.
- [17] Han Liang; Ye Yizheng; Bai Tao, "2.5Gb/s CMOS RGC transimpedance amplifier," *In Proc. Of International Conference on Microelectronics, ICM*, vol.4, no.2., pp.19-22, 29-31 Dec. 2007.